

I. Corporate Employee Self-Service (ESS) Portal

Complete Project Documentation

Employee Portal:

1. Project Goal:

The goal of this document is to build a simplified version of the Corporate Employee Self-Service Portal. Instead of launching all features at once, we are focusing strictly on automating basic HR workflows like recording daily attendance, managing leave applications, and securely viewing payslips. This will reduce manual HR paperwork and give employees immediate access to their core data.

2. Core User Roles:

The roles from the original documentation to just three essential types:

Employee: Can check in/out, submit leave requests, and see their own payslips.

Manager: Can do everything an employee can do, plus view and approve/deny leave requests

HR / Admin: Can manage the database, upload PDF payslips, view attendance.

3. Feature Scope:

Basic Profile & Login: Secure authentication using simple email and password logins.

Attendance Tracking: A digital web button for "Clock In" and "Clock Out" that logs exact timestamps.

Leave Management: A basic online form for employees to choose dates and submit a leave request. With approve and reject for Managers.

Payslip Viewer: An admin portal where HR can upload individual PDF payslips

4.Simplified Database Schema:

We will start with only 5 main database tables instead of the full list:

Employees Table: Stores basic profiles (ID, Name, Email, Role, Department).

Departments Table: Simple mapping of department names.

Attendance Table: Logs user ID, date, clock-in time, and clock-out time.

Leaves Table: Tracks leave type, start/end dates, status

Payslips Table: Stores links to uploaded PDF files mapped to specific employee IDs and months

5. Technology Stack:

We will use required tech stack based on the data and the client's feedback.
We will opt these:

Frontend: Next.js with Tailwind CSS

Backend: Node.js with Express.

Database: SQL

File Storage

II. Internship Management, Collaboration & Mentorship Portal

Internship Student-Admin Portal:

1. Project Goal:

This internship portal is mainly for internship cohort. They can access the allotted tasks easily from their respective mentors. Tracking the progress of the intern and finalizing the certificate becomes much easy.

2. Core User Roles:

Administrator: Handles system configuration, user onboarding, and matching mentors to interns

Mentor: Assigns daily/weekly tasks, monitors progress, and completes the final performance evaluation

Intern: Receives tasks, submits daily progress updates, and downloads their internship certificate upon completion

3. Feature Scope:

Standard Login: Secure email and password authentication

Task Creation & Assignment: Mentors can create tasks with a title, description, priority, and deadline, assigning them directly to specific interns.

Mentor Review: Reviews all the tasks and tells the intern if any mistakes.

4. Technology Stack:

Frontend: React.js, Next.js, and TypeScript

Backend: Node.js and Express.js

Database: SQL

III. Video Communication & Virtual Collaboration

Platform Documentation

1. Project Goal:

To deliver a stable, high-performance web-based video communication platform that allows users to seamlessly create, invite others to, and participate in reliable, real-time video meetings without the overhead of enterprise infrastructure.

2. Core User Roles:

Host : A user who creates an account, starts instant meetings

Guest : A user who joins an active meeting

3. Feature Scope:

Registration & Login: Standard email and password authentication

Password Recovery: Basic "Forgot Password" email reset flow.

Instant Meetings: Users can create a meeting room immediately with a single click

Meeting Invitations: Generation of a unique, shareable URL that allows guests to join the meeting

Screen Sharing: Ability for a single presenter to share their entire screen or a specific window

Video & Audio Engine Gallery/Speaker views, standard mute/unmute toggles, and audio device selection.

4. Technology Stack:

Frontend UI: React.js

Backend Application Server: Node.js

Media Streaming Engine: WebRTC)

Primary Database: PostgreSQL

IV. Enterprise Work Management & Collaboration Platform - Complete Product Documentation

1. Project Goal:

The primary objective of the MVP is to build a lean, functional work management platform that proves the core utility loop: allowing teams to create projects, organize tasks visually, and collaborate in real-time.

2. Core User Roles:

Workspace Admin: Has full permissions to create workspaces, start new projects, invite team members

Team Member: Has permission to view assigned projects, create tasks, edit task details

3. Feature Scope:

Onboarding: Basic user registration, secure login, and password recovery

Project Creation: Manual creation of project workspaces

Task Engine: Standard capability to create, edit, view, and delete tasks

Basic Collaboration: Contextual comment threads within individual tasks

4. Technology Stack:

Frontend: React.js, TypeScript, Redux.

Backend: Node.js, Express.js.

Database: PostgreSQL

Real-time: Socket.IO

5. Database Schema:

Users Table:(id, email, password ,role ,created_date)

Projects Table: (id, name,, created_by, created_at)

Tasks Table : (id, project , title , status, due_date)

V. Microsoft Teams Replica - Complete Project Documentation

1. Project Goal:

To build a functional, real-time enterprise collaboration platform that enables users to securely communicate via 1-to-1 chats, group chats, and organized channels while sharing files.

2. Core User Roles:

System Admin: Responsible for basic user management, team oversight, and system monitoring.

Team Owner: A user who can create public/private teams, manage member invitations, and configure channels

Team Member: A standard employee who can chat, join channels, send direct messages, and upload/download files

3. Feature Scope:

Basic Authentication: User registration, secure login, and profile creation.

Personal Chat (1-to1): Real-time text messaging, typing indicators

Teams & Channels: Ability to create Public/Private Teams

Basic File Sharing: Uploading, downloading, and previewing images or documents within chats and channels.

Audio/Video Calling: One-to-one calls, scheduled group meetings, screen sharing, and recording

4. Technology Stack:

Frontend: React.js

Backend: Node.js with Express

Realtime Engine: Socket.IO

Database: MongoDB